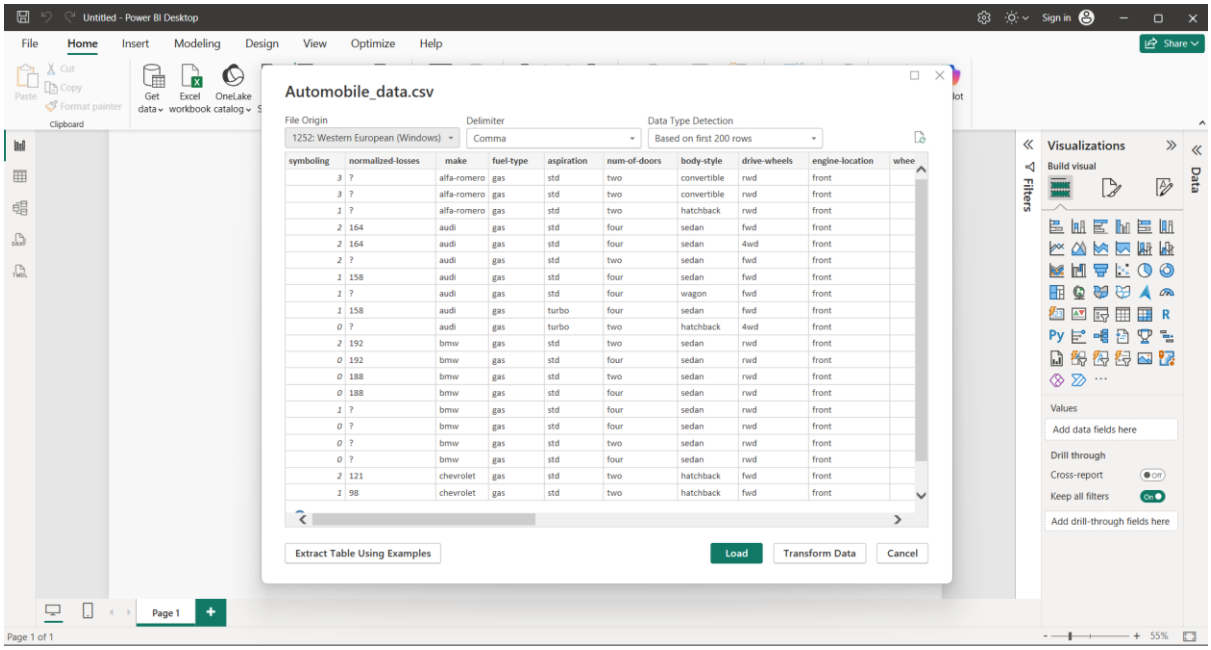
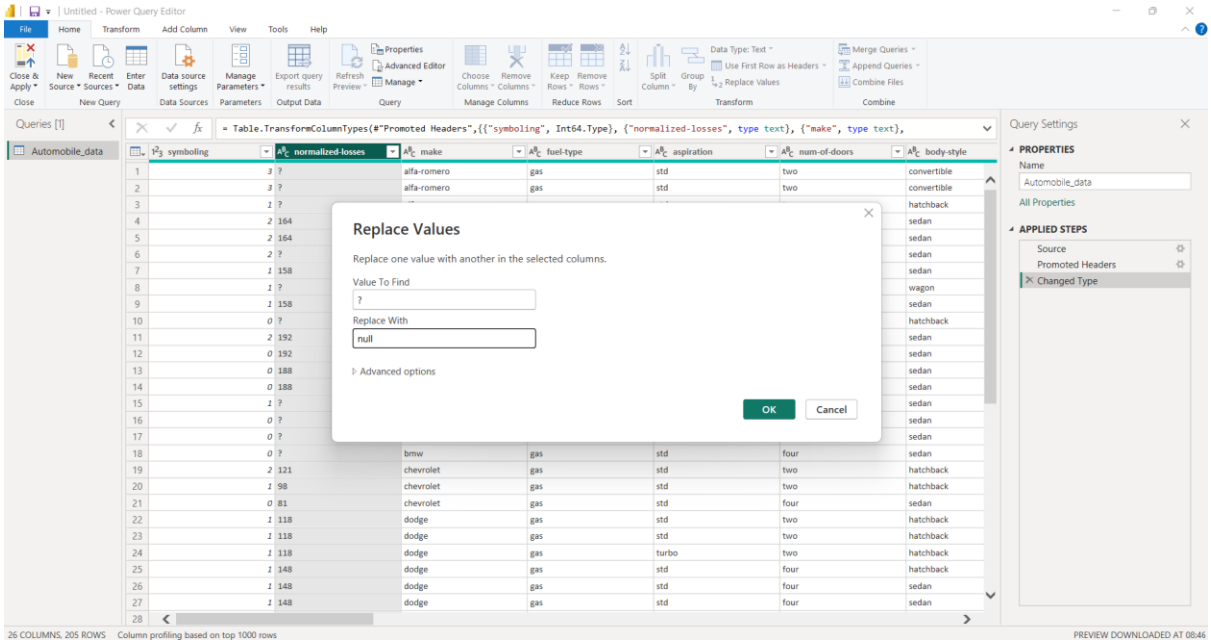
WEEK 11

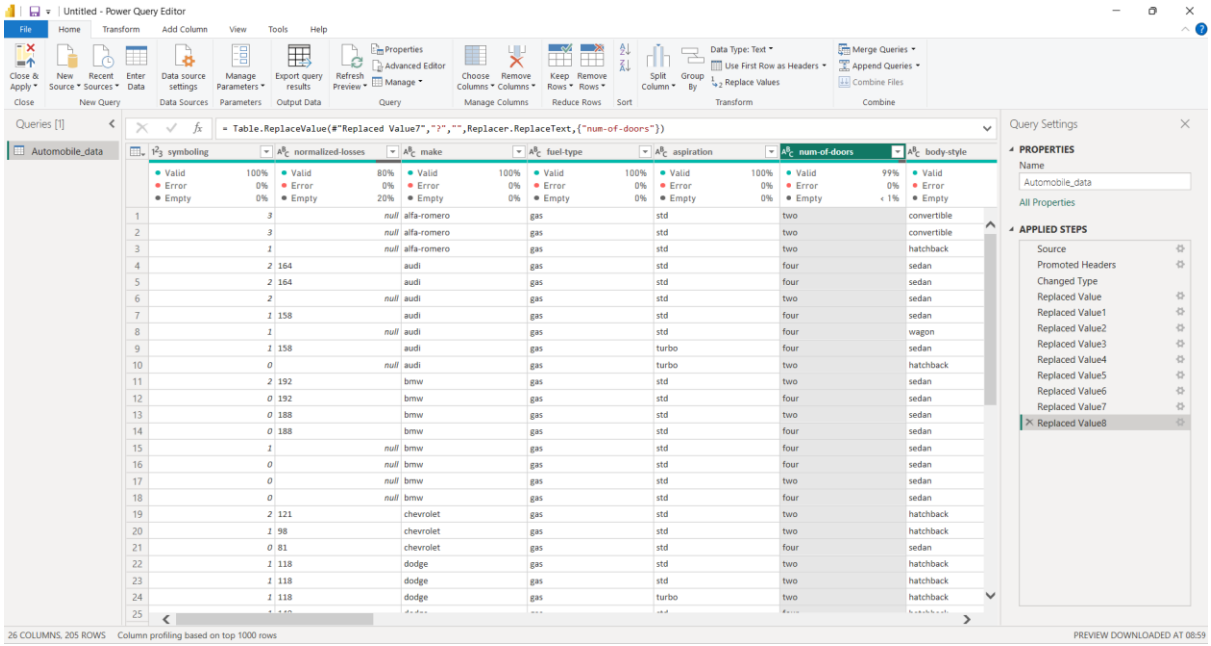
SAI SHRUTI B



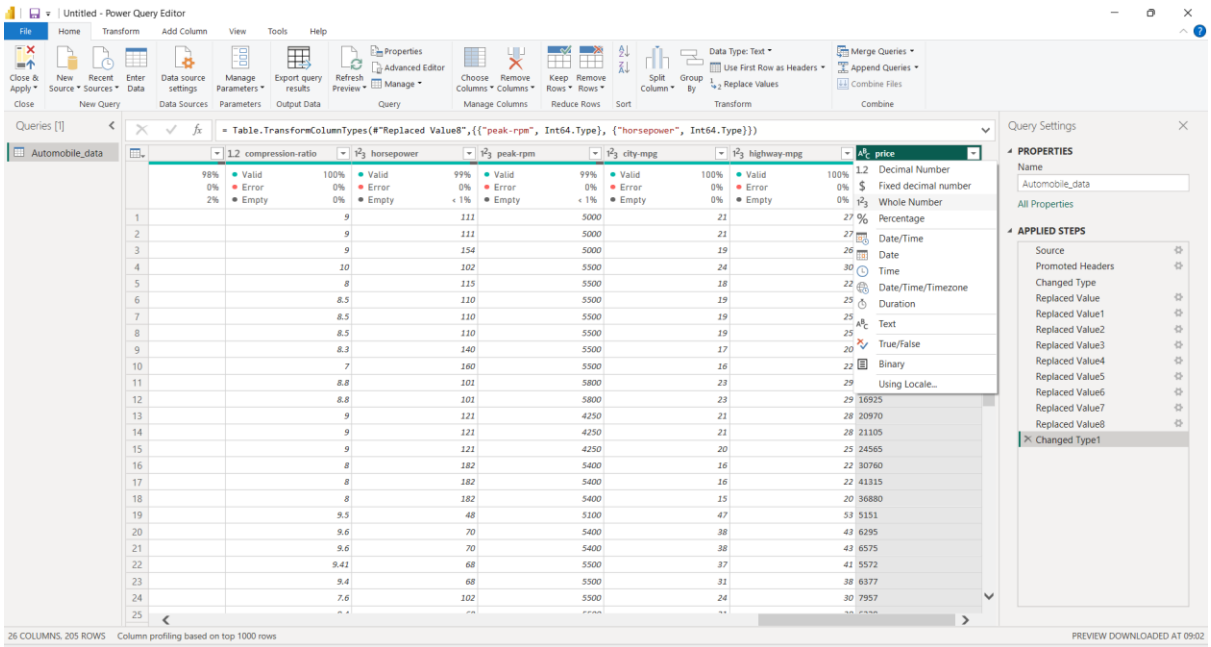
The Automobile_data.csv dataset was imported into Power BI using the Text/CSV option. The dataset was previewed before loading it into Power Query for further cleaning and transformation.



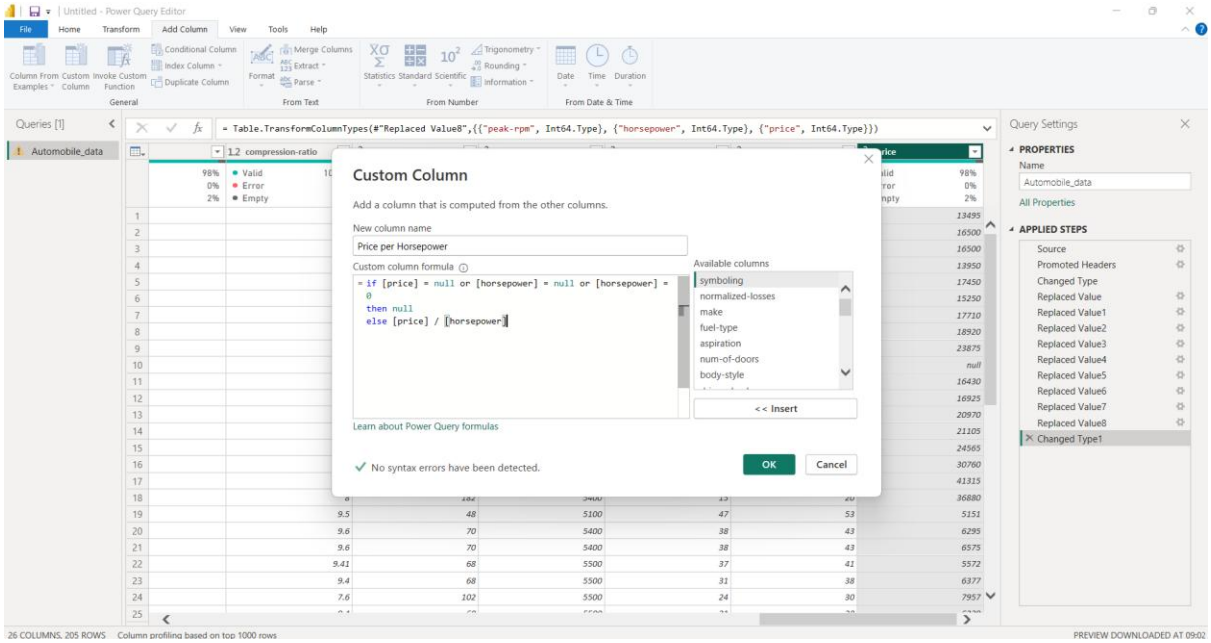
The Replace Values option was used in Power Query to replace the invalid ? placeholders with null values in the dataset.



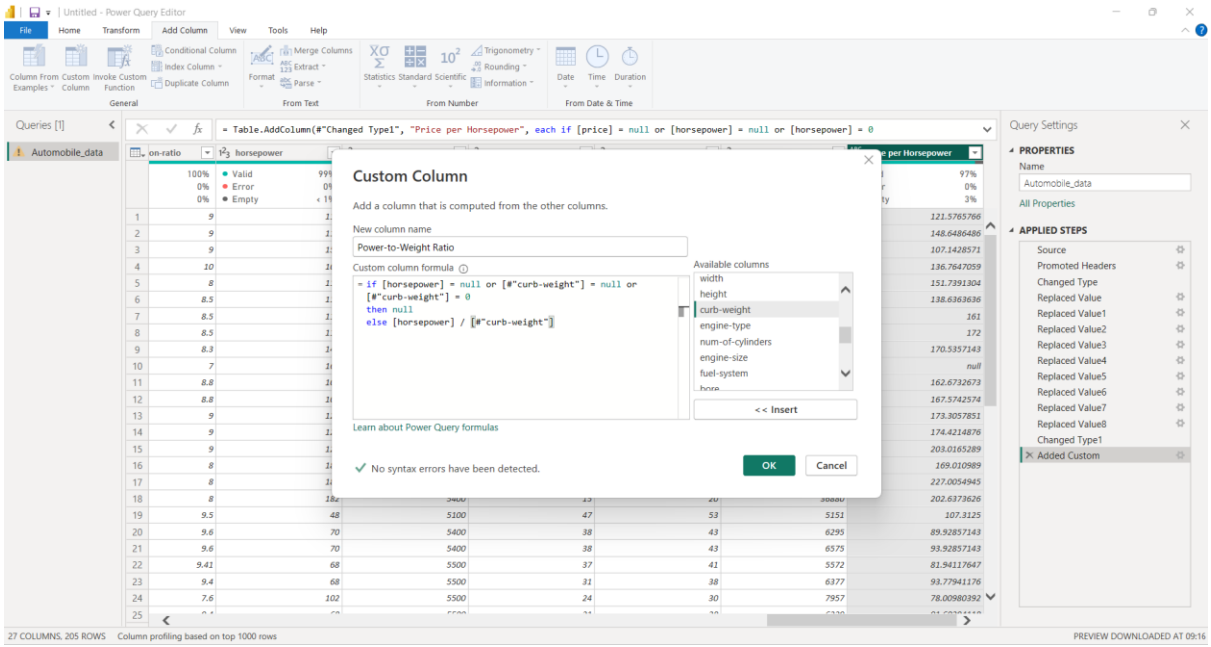
The replacement operation was applied to the relevant columns. The invalid placeholders were converted into null values, making the dataset suitable for further processing.



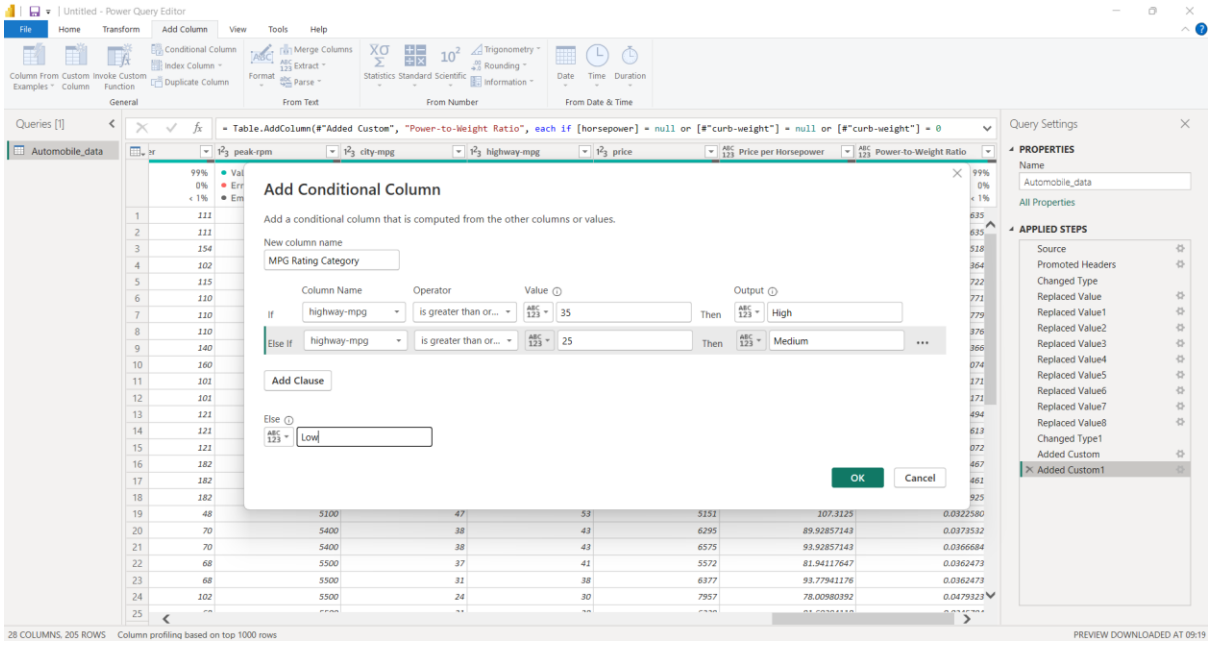
The data types of the horsepower, peak-rpm, and price columns were changed to numeric types so that mathematical calculations could be performed on them.



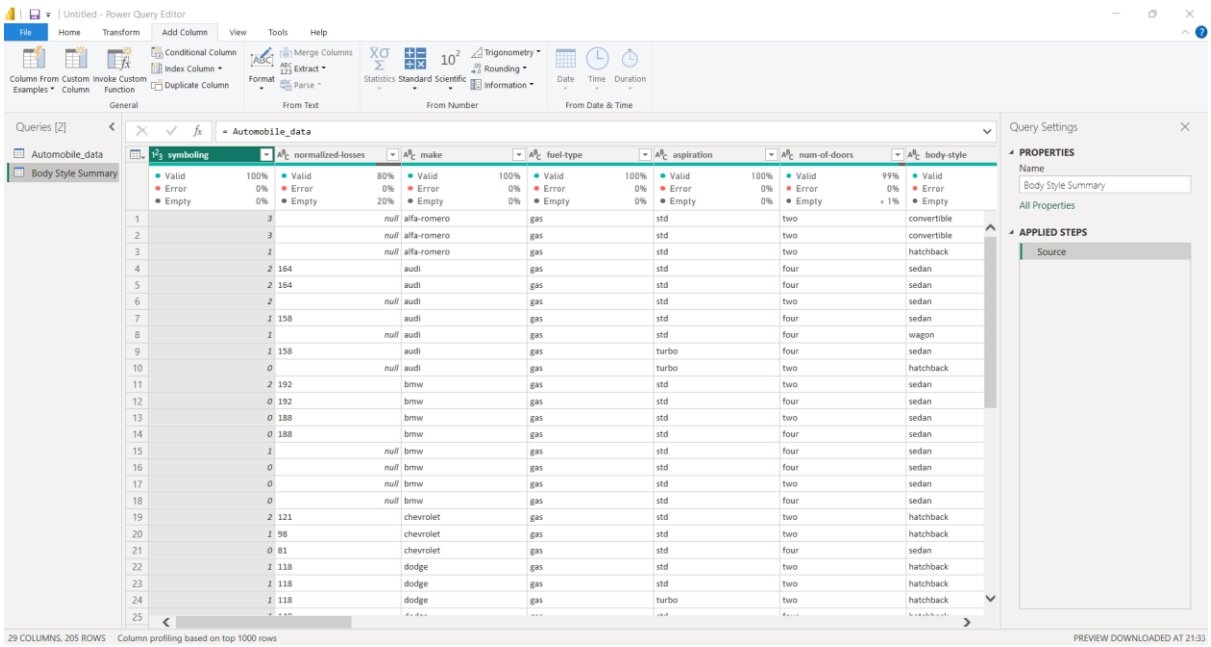
A custom column named Price per Horsepower was created using the vehicle's price and horsepower. The calculation also checks for null or zero horsepower values to avoid invalid results.



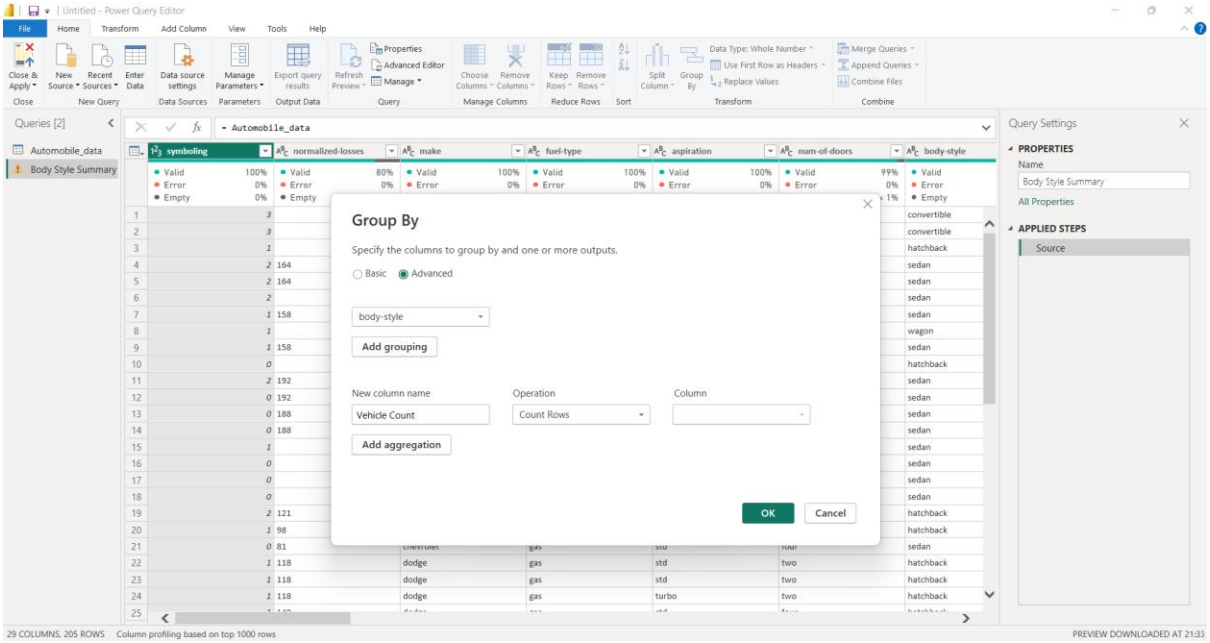
A custom column named Power-to-Weight Ratio was added by dividing horsepower by curb weight. This represents the horsepower available relative to the vehicle's weight.



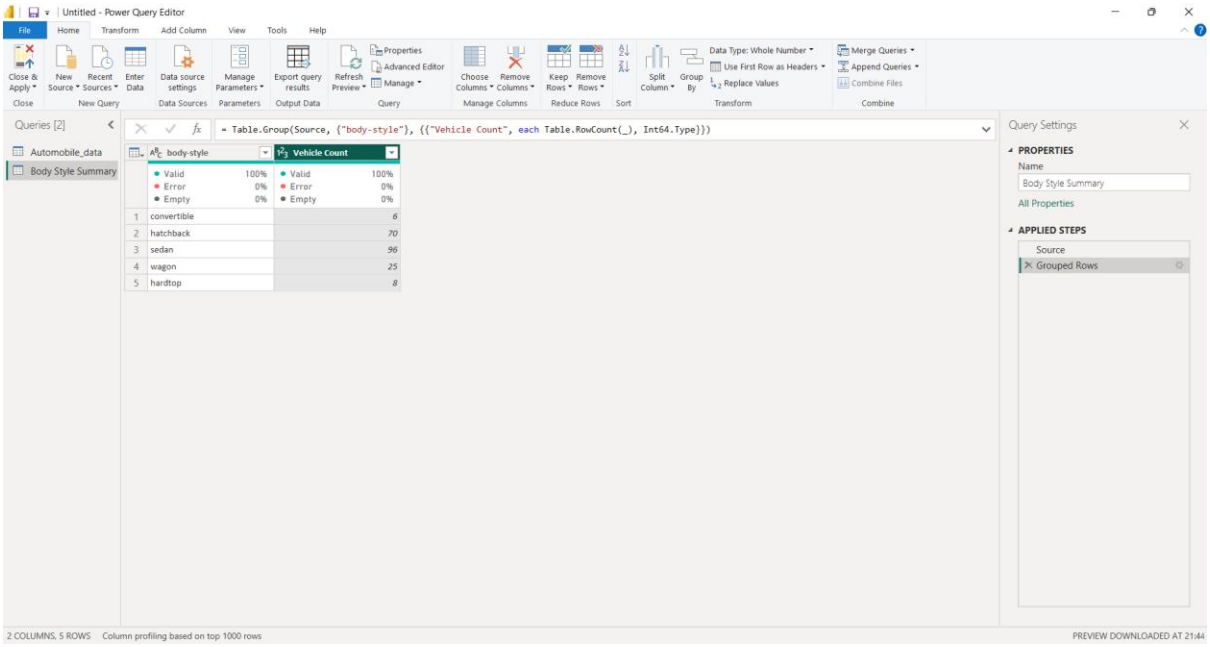
A conditional column named MPG Rating Category was created using highway MPG values. Vehicles were classified as High, Medium, or Low based on the selected mileage ranges.



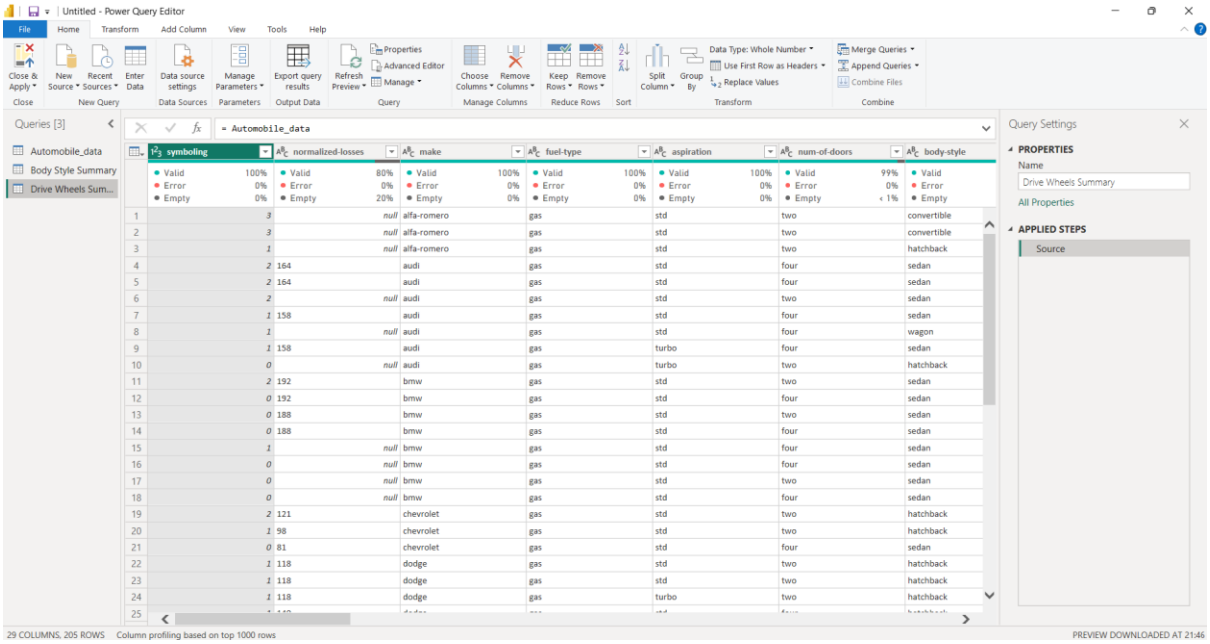
The Power Query preview displays the newly created derived columns, including Price per Horsepower, Power-to-Weight Ratio, and MPG Rating Category.



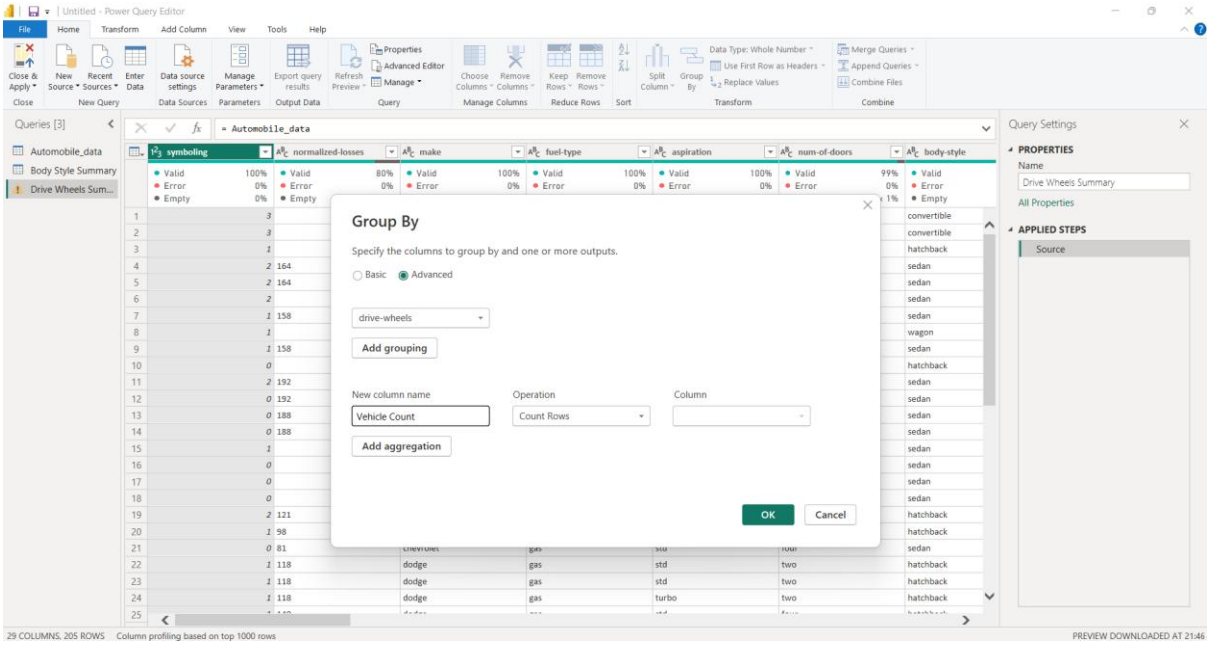
The Group By option was configured using the body-style column. The Count Rows operation was selected to calculate the number of vehicles in each body-style category.



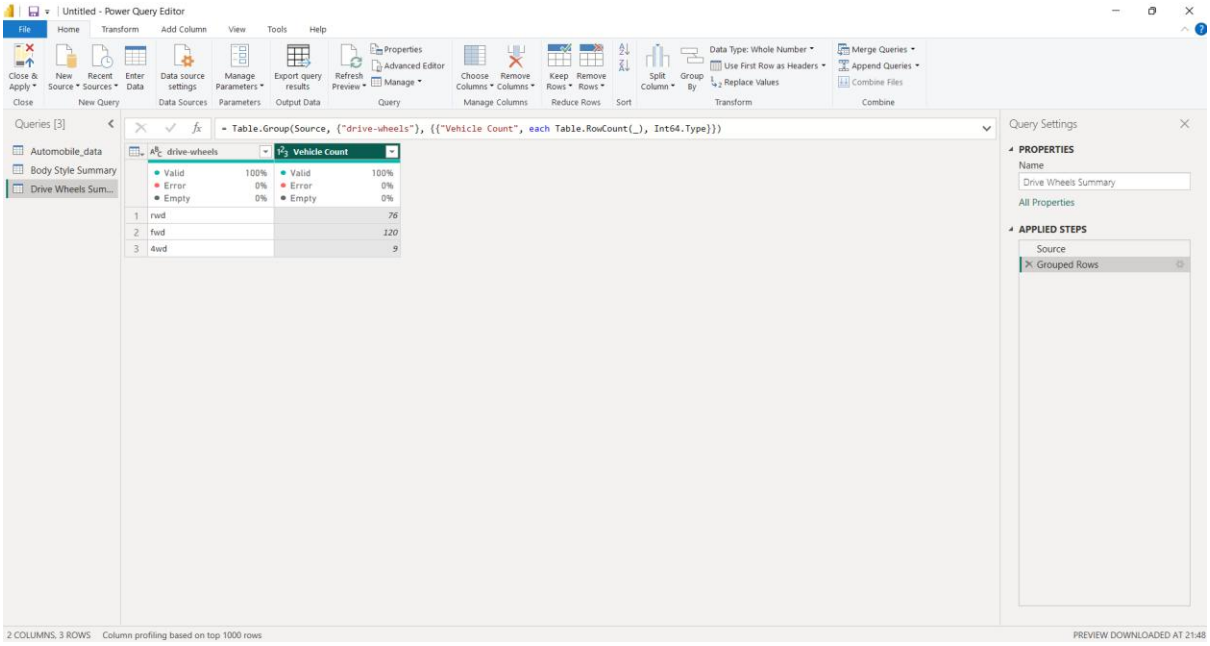
The resulting summary displays the vehicle count for each body style, including convertible, hatchback, sedan, wagon, and hardtop.



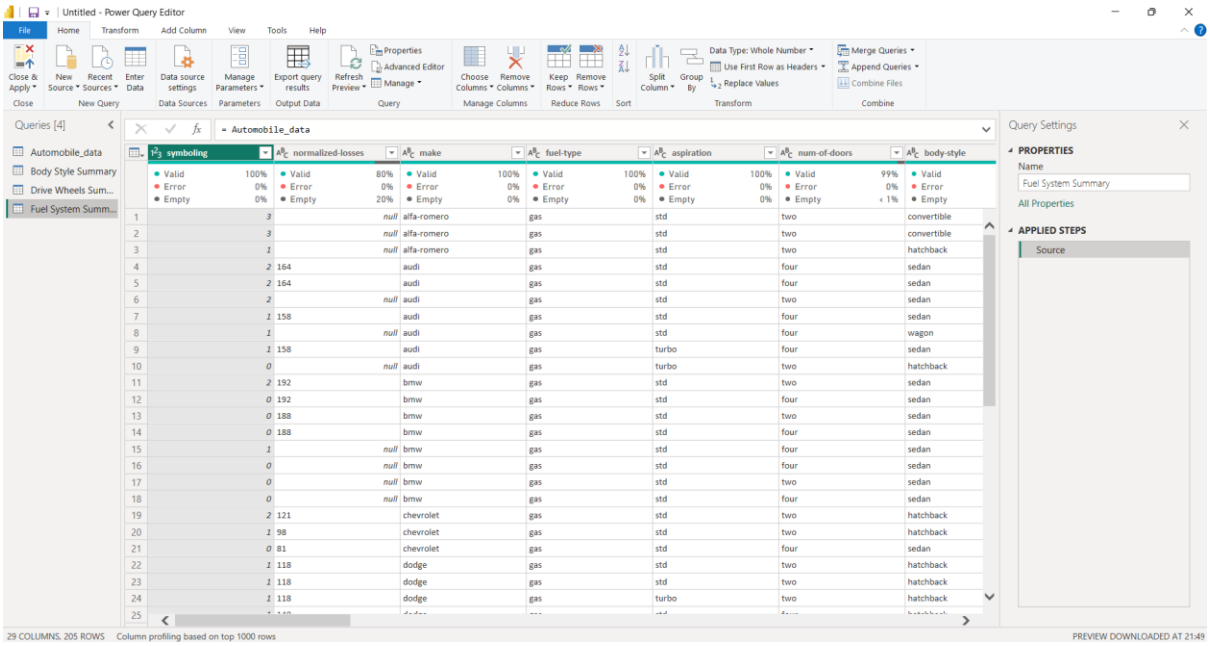
A referenced query named Drive Wheels Summary was created from the original automobile dataset to perform a separate grouping operation.



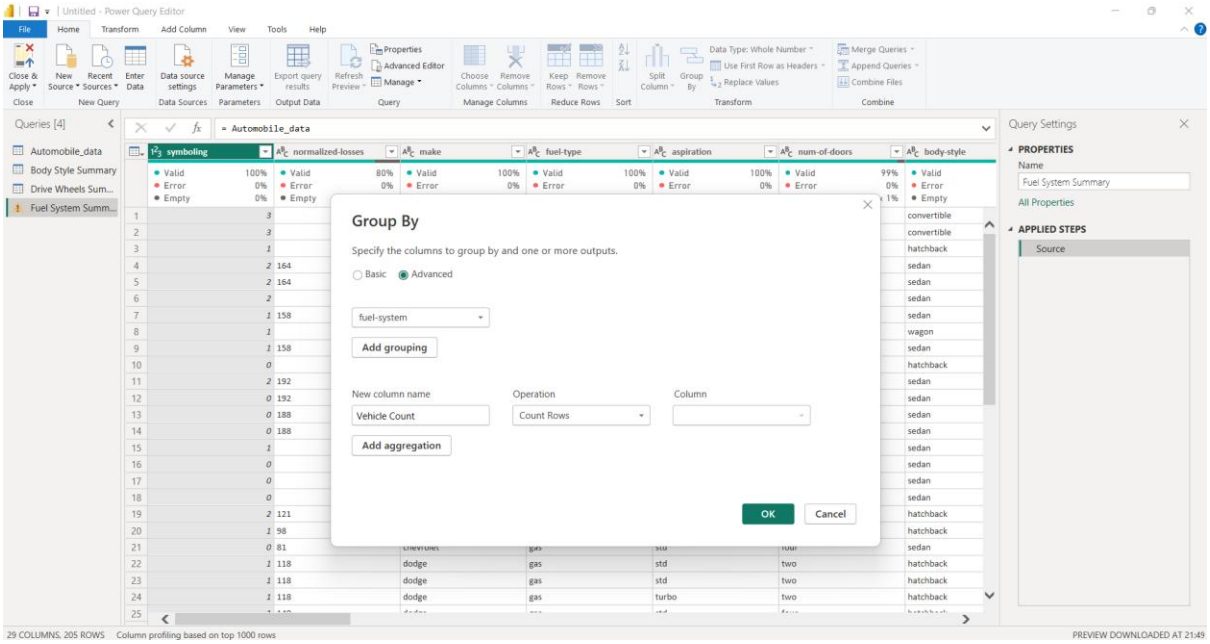
The Group By option was configured using the drive-wheels column, with Count Rows selected to calculate the number of vehicles in each category.



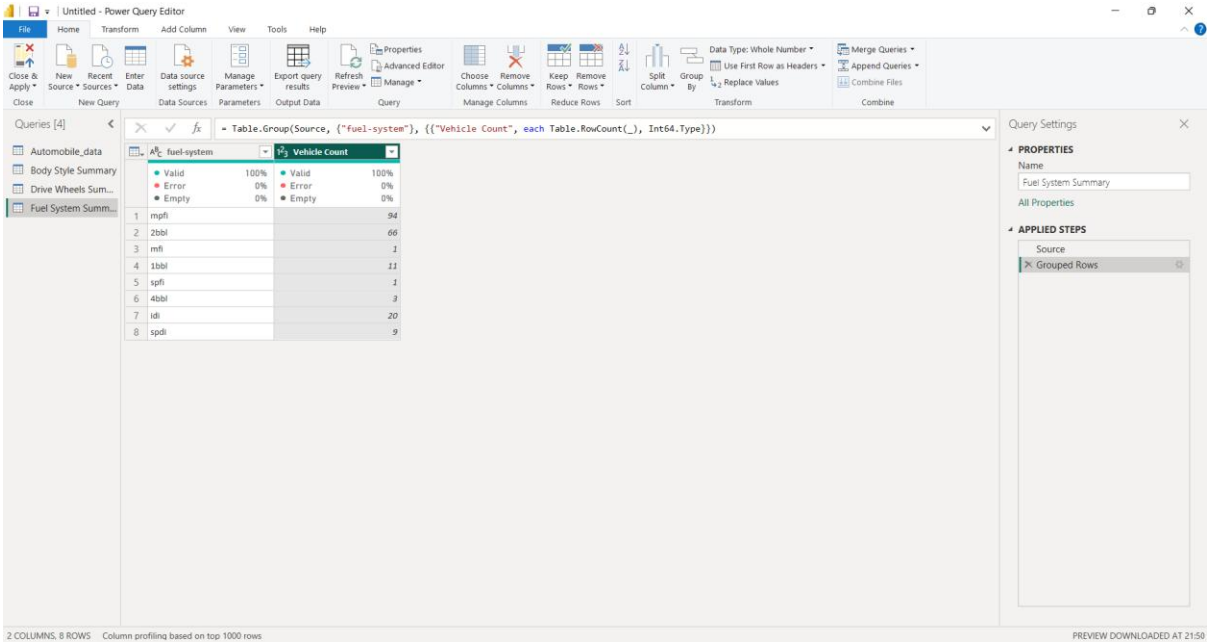
The grouped result displays the number of vehicles with rear-wheel drive (RWD), front-wheel drive (FWD), and four-wheel drive (4WD).



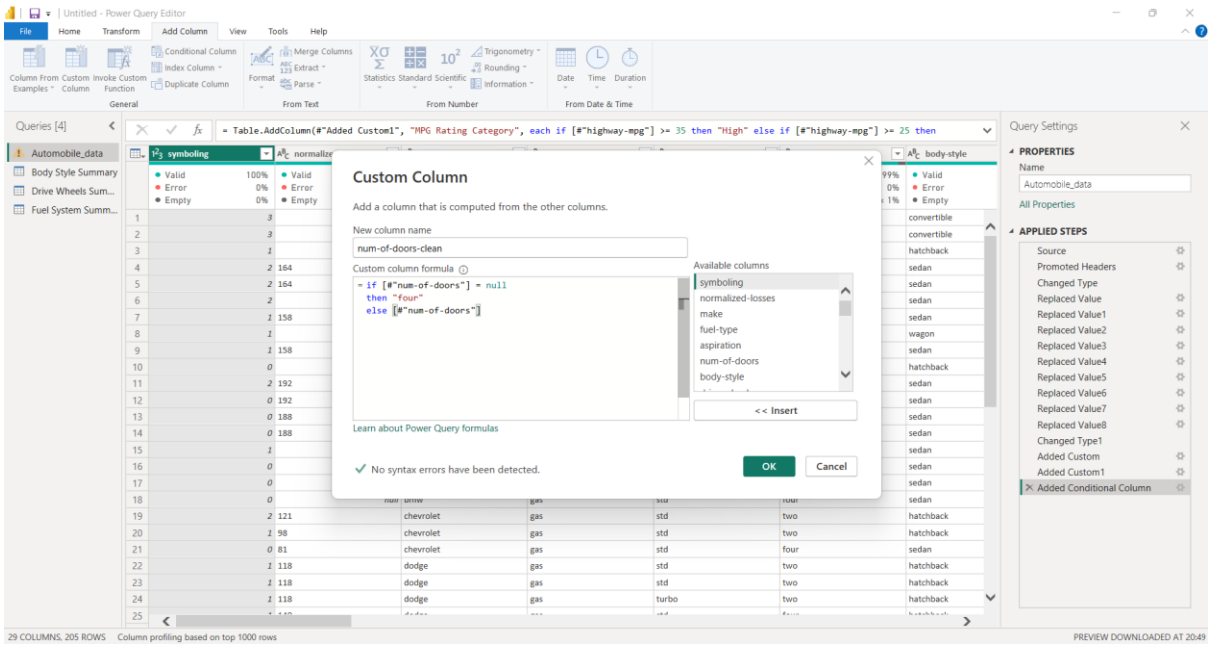
A separate referenced query named Fuel System Summary was created from the original automobile dataset for fuel-system analysis.



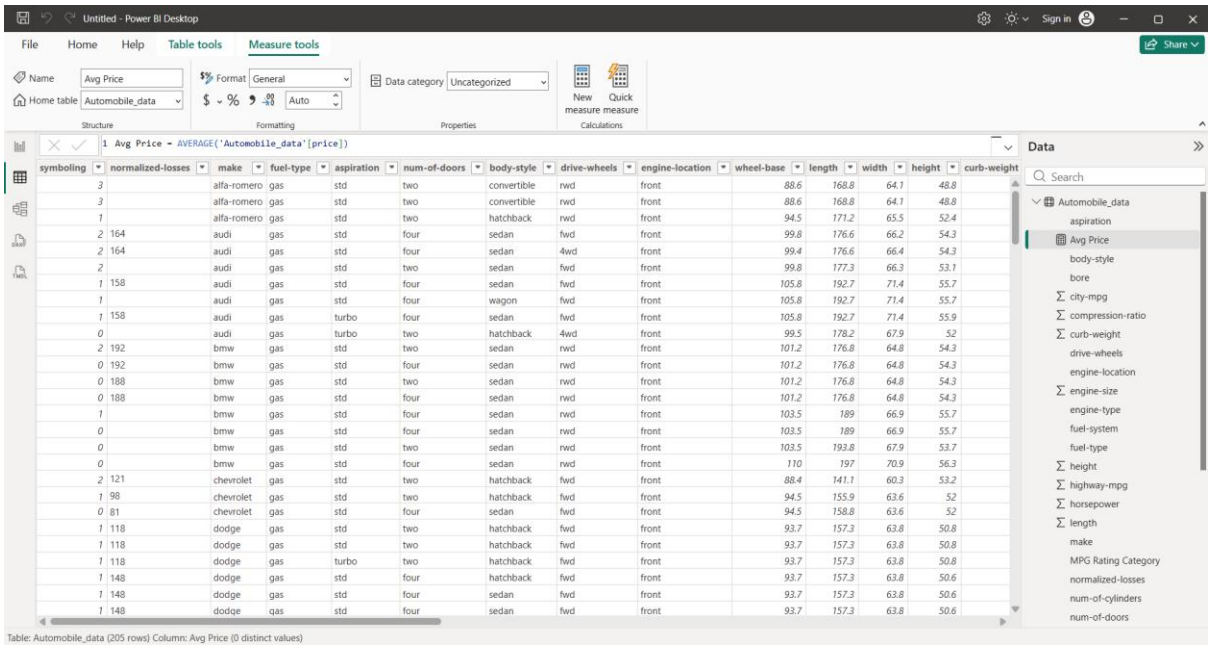
The Group By operation was configured using the fuel-system column, with Count Rows selected as the aggregation.



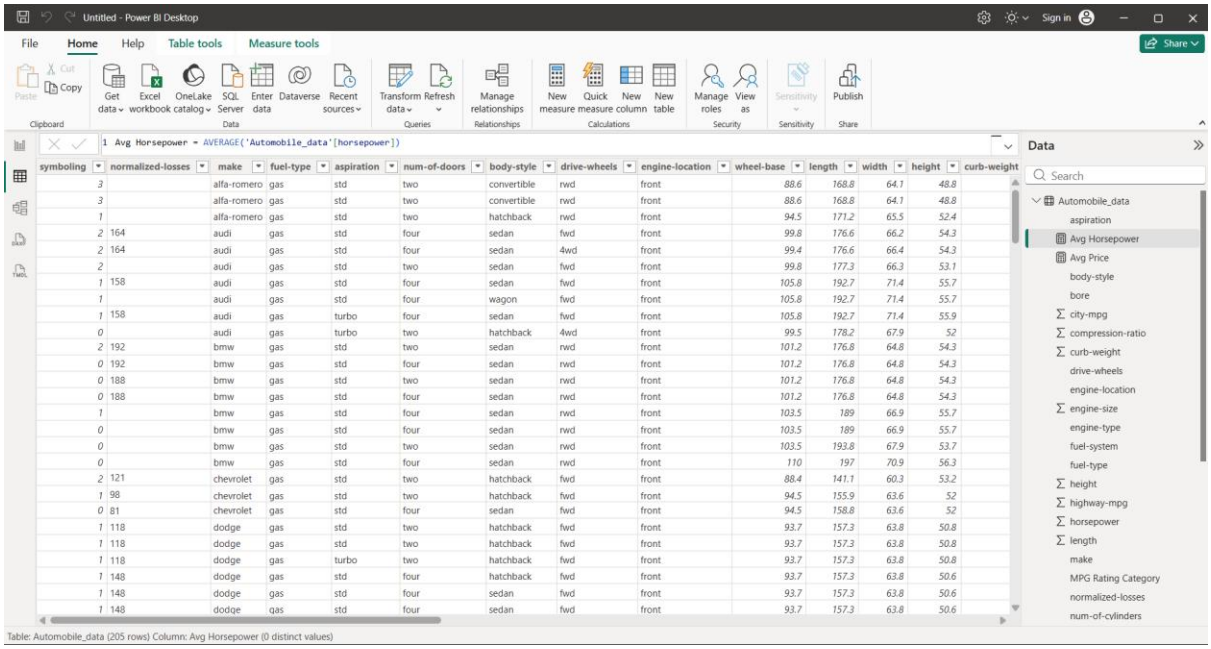
The resulting table displays the vehicle count for each fuel-system category, providing a summary of the different fuel systems present in the dataset.



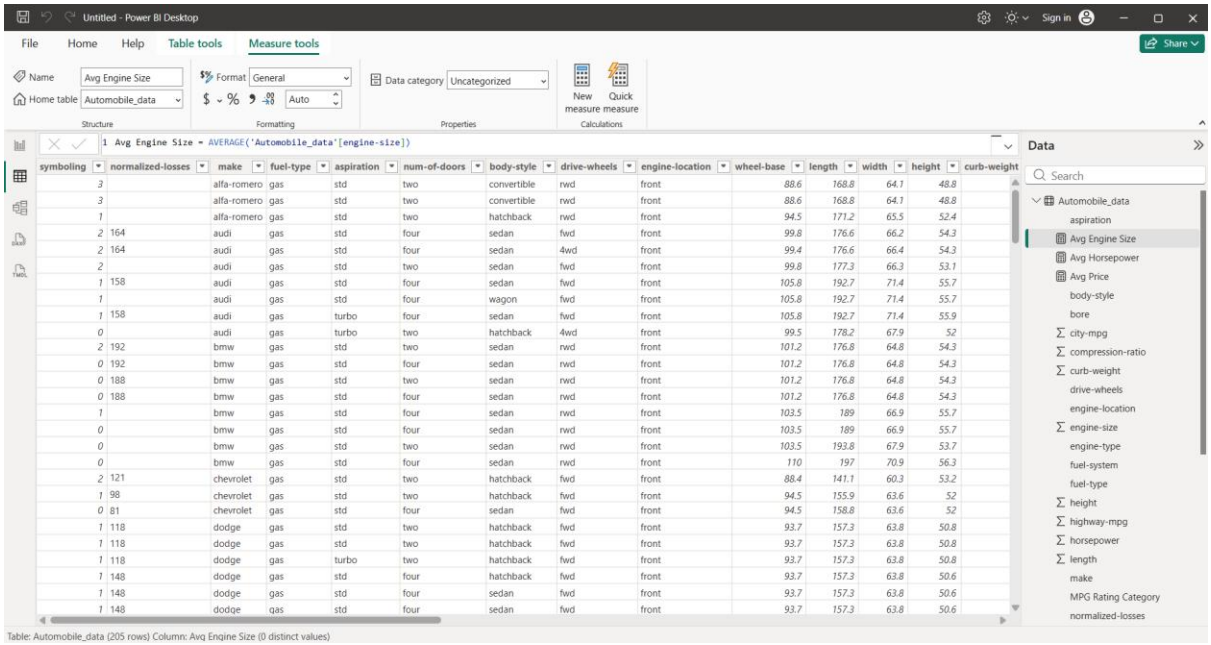
A custom column named num-of-doors-clean was created using conditional logic. Missing values in the num-of-doors column were replaced with four, while existing values were retained.



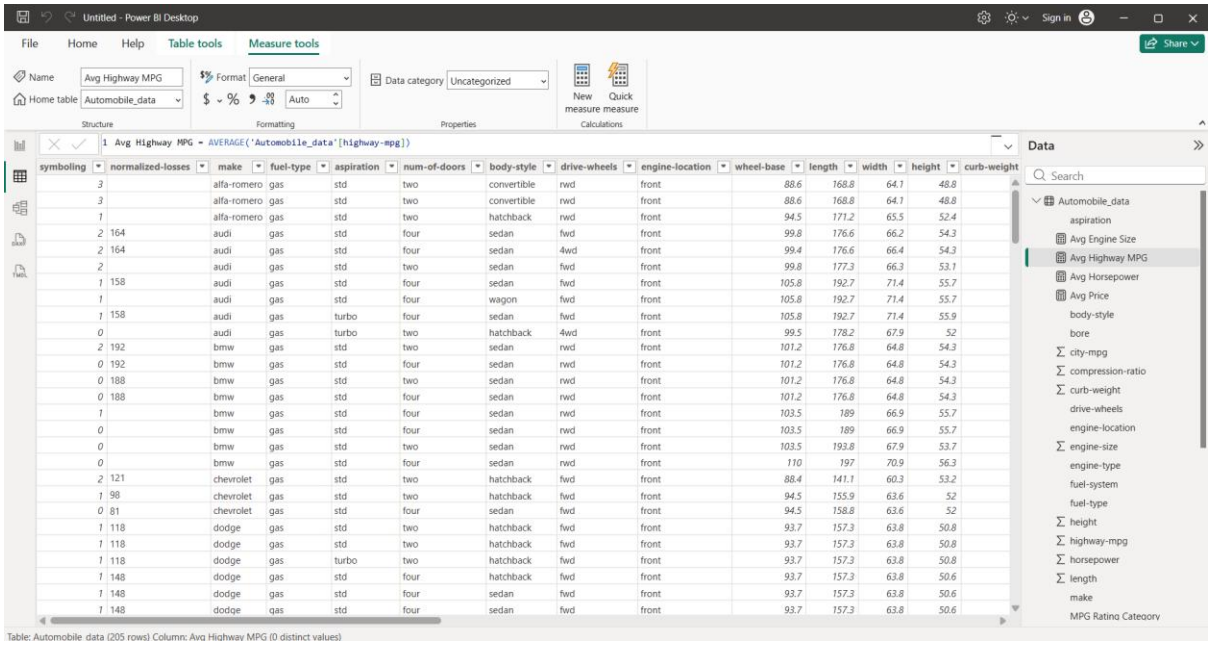
A DAX measure named Avg Price was created using the AVERAGE function to calculate the average vehicle price.



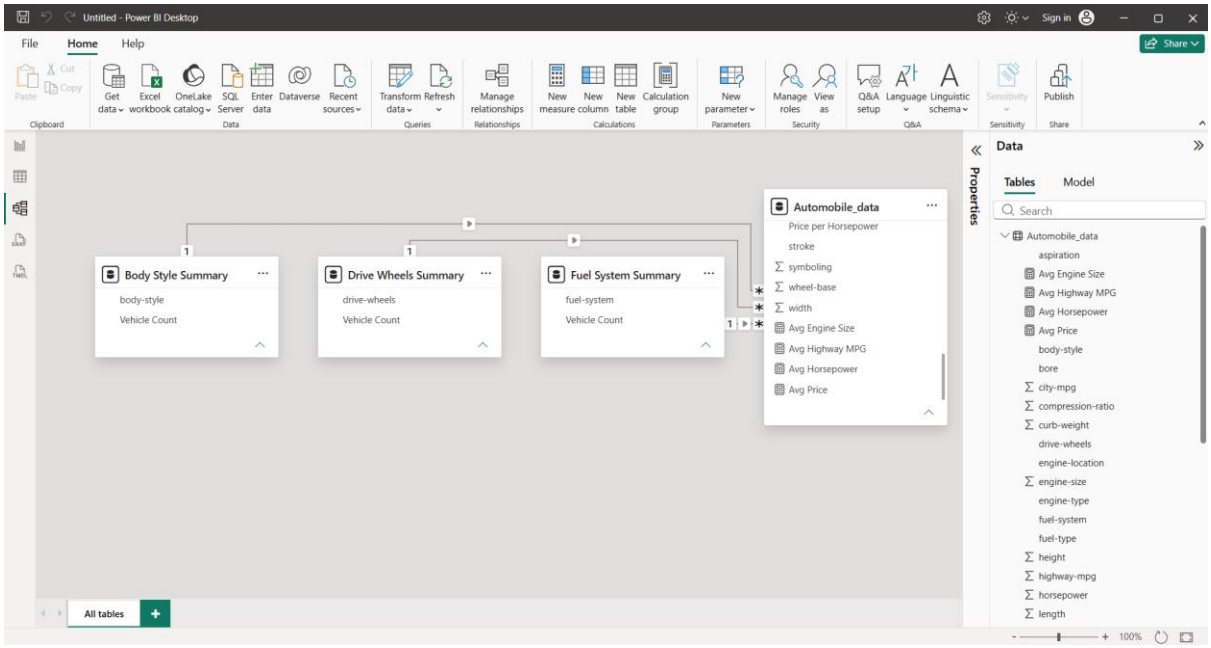
A DAX measure named Avg Horsepower was created using the AVERAGE function to calculate the average horsepower of the vehicles.



A DAX measure named Avg Engine Size was created to calculate the average engine size across the automobile dataset.



A DAX measure named Avg Highway MPG was created using the AVERAGE function to calculate the average highway mileage.



The Model View displays the original Automobile_data table along with the three summary tables: Body Style Summary, Drive Wheels Summary, and Fuel System Summary. This provides an overview of the tables created during the data preparation process.